A3-PS021-TC509

SENTRY-6G

Failure-Aware Dialog Continuity and Safe Recovery Across Heterogeneous AI-Agent Networks

PS-021 | Experimental Dialog Correlation and Recovery Across Agent Adapters

Shriyam Adya Sharma

Email: shriyamadyas.cs23@rvce.edu.in

Shreyas G

Email: shreyasg.cs23@rvce.edu.in

AIORI Hackathon 3

14 October 2026

Problem Description

A task may finish at Agent B while Agent A sees only a timeout: did the action fail, or was its acknowledgment lost?

01	**Dialog identity breaks**	Different adapter IDs and restarts may sever a single logical task, causing an unintended new request.
02	**Execution becomes uncertain**	A network-control command may commit at Agent B while its acknowledgment never reaches Agent A.
03	**Retries may repeat effects**	Duplicate/concurrent requests can replay non-idempotent actions; late updates may overwrite valid states.
04	**Recovery cannot be assumed**	Disconnections and crashes require durable correlation, validated state changes and evidence before resuming.

Background

Why an agent-to-agent timeout is more than a lost message

Heterogeneous adapters

Independent agents may use different task IDs, states and protocols; a dialog requires explicit correlation.

Unreliable networks

Loss, reordering, partitions and crashes can hide whether the delegated action actually executed.

Safety and continuity

Persistent state, idempotency and verified execution evidence support safe retries and resumption.

Network-ops example

Adapter B changes a router setting; the reply is lost. A must determine whether it committed before any unsafe replay (RFC 9110 [6]).

Literature & RFC Review

Verified IEEE research and standards establish agent coordination, recovery and retry foundations.

SOURCE / YEAR	ESTABLISHED CONTRIBUTION	RELEVANCE / LIMIT
IEEE IRI · 2026 [1]	Event correlation; deferred context reconciliation; SLA-aware recovery	Closest recovery-related work
IEEE TCCN · 2025 [2]	Internet of Agents; orchestration and heterogeneous collaboration	Agent-network foundations
IEEE Commun. Mag. · 2026 [3]	Agentic-AI network architecture, protocols and 6G context	Future-network motivation
IEEE EdgeCom · 2025 [4]	A2A communications at the network edge; protocol limitations	Edge-network constraints
A2A specification [5]	Task IDs, context IDs, lifecycle and status retrieval	Existing protocol capabilities
RFC 9110 · 2022 [6]	HTTP idempotency and retry semantics (§9.2.2)	Safety principle; foundational

Research question: can evidence-gated, network-aware reconciliation improve verified recovery time or overhead over durable idempotent recovery?

Proposed Solution

A prototype recovery layer for two mock agent adapters—with a testable, safety-first policy.

1	**Correlate**	Assign one global dialog ID; map both local task IDs and enforce lifecycle transitions.
2	**Persist**	Persist dialog journal and outbox; enforce atomic idempotency at the executor effect boundary.
3	**Classify**	Use authoritative executor evidence: VERIFIED DONE / VERIFIED NOT EXECUTED / UNKNOWN.
4	**Recover**	Return prior result; safely retry; query, wait or escalate. Never blindly replay unknown effects.
5	**Adapt**	For permitted options only, choose query/backoff timing using observed network conditions.

Live fault case: Agent B commits router change OP-501 → acknowledgment lost → SENTRY verifies executor ledger → returns original result; no second effect.

Optimization Proposed

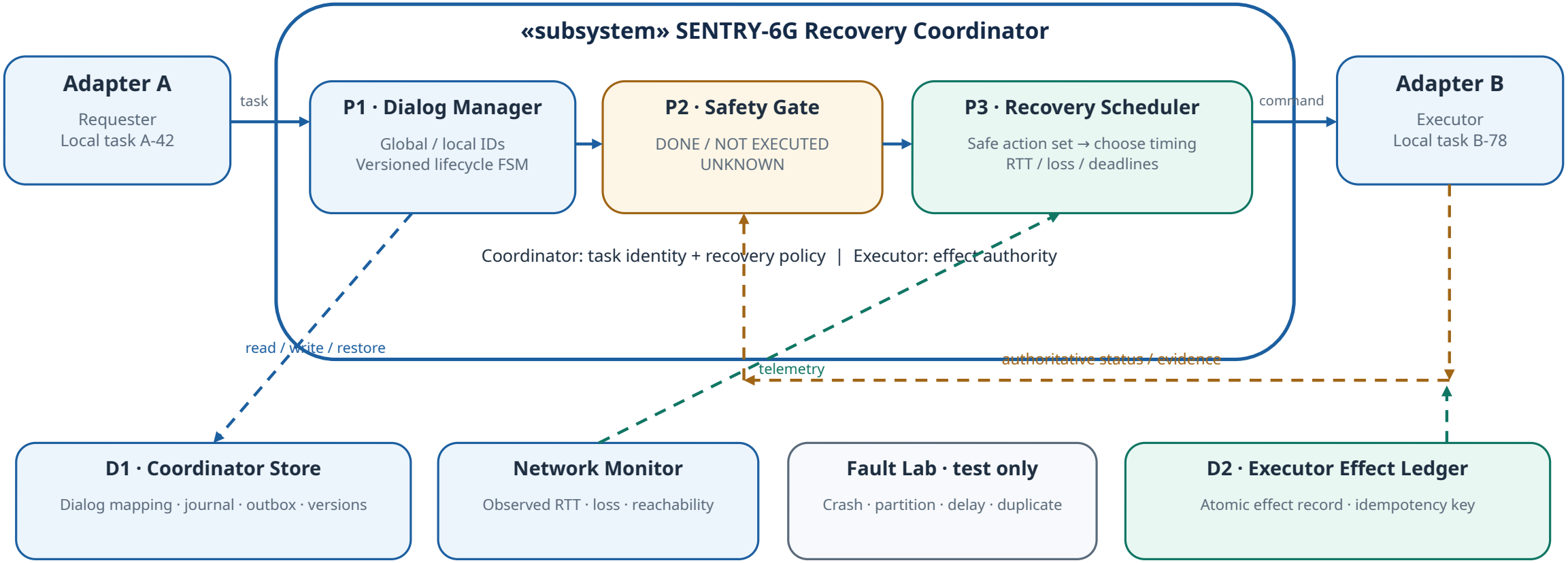
Comparison against a credible durable-recovery baseline (B2) — not measured superiority

EVALUATION DIMENSION	DURABLE BASELINE (B2)	SENTRY-6G HYPOTHESIS
Uncertain outcome	Executor idempotency + status reconciliation	Classify evidence; gate each permitted action
Impaired network	Fixed polling or exponential backoff	Adapt status-query timing to measured RTT/loss
Experimental evidence	Durable recovery and fault-test results	Compare safety, latency, messages, unresolved tasks

What we must prove: lower time-to-safe-resolution or messaging cost than B2, with no additional duplicate effects.

Solution Architecture & Design

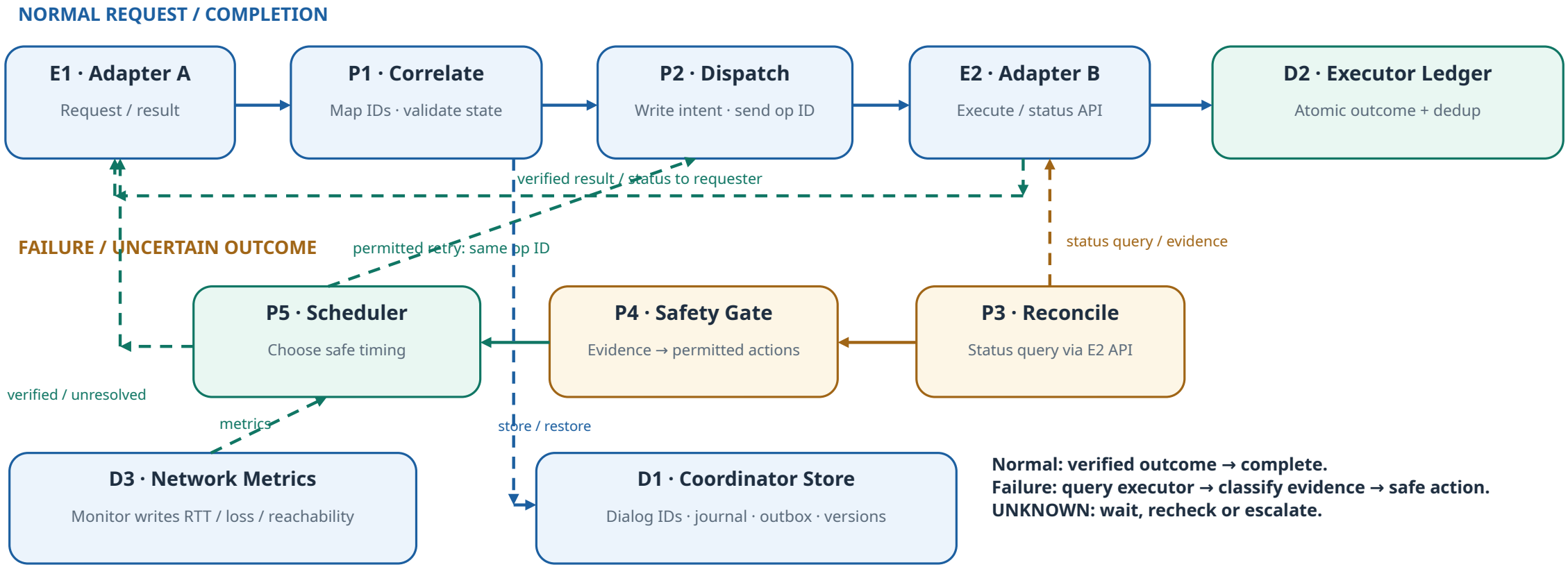
UML-style component view • Request/command APIs (solid) • Evidence, persistence and telemetry (dashed)



INVARIANT Safety checks precede network optimization; retries require an executor-enforced idempotency contract.

Data Flow Diagram

Level-1 data-flow diagram | E = external entity P = process D = data store • Data flows, not execution order



Executor ledger is accessed through Adapter B, never directly by the coordinator. No blind replay of uncertain effects.

Deployment Timeline

Milestone-driven development: official compliance first, comparative research second.

Phase 1	**Level 1 · Proposal / design**	14 Oct: repo, schema v0.1, lifecycle table, architecture and submission PDF.
Phase 2	**Two mock adapters + persistence**	16–23 Oct: global/local mapping, journal, atomic dedup and five scripted faults.
Phase 3	**Recovery controller + fault lab**	24–31 Oct: evidence classification, reconnect policy, partitions and dashboard.
Phase 4	**Validation & mentor demo**	Nov: B2 baseline, repeated experiments, overhead/latency metrics and ablations.

Demo exit criteria: five reproducible faults; stable dialog ID after restart; one committed effect; explicit UNKNOWN; B2 comparison planned.

Bibliography

[1] S. R. K. V. Bayyavarapu et al., “Event-Correlated MCP Agents for Asynchronous Enterprise Systems,” IEEE IRI, 2026. DOI: 10.1109/IRI69576.2026.00011.

[2] “Internet of Agents: Fundamentals, Applications, and Challenges,” IEEE Transactions on Cognitive Communications and Networking, 2025. DOI: 10.1109/TCCN.2025.3623369.

[3] “Toward the Internet of Agentic AI: Protocols, Architecture, and Challenges,” IEEE Communications Magazine, 2026. DOI: 10.1109/MCOM.001.2500309.

[4] Q. Duan and Z. Lu, “Agent Communications toward Agentic AI at Edge—A Case Study of the Agent2Agent Protocol,” IEEE EdgeCom, 2025. DOI: 10.1109/EdgeCom66327.2025.00032.

[5] A2A Project, “Agent2Agent (A2A) Protocol Specification,” official specification. https://a2a-protocol.org/v1.0.0/

[6] R. Fielding, M. Nottingham, and J. Reschke, “HTTP Semantics,” RFC 9110, §9.2.2, 2022. https://www.rfc-editor.org/rfc/rfc9110

[7] “Assessing and Enhancing the Robustness of LLM-Based Multi-Agent Systems Through Chaos Engineering,” IEEE/ACM CAIN, 2025. DOI: 10.1109/CAIN66642.2025.00039.

Thank You for Your Attention!

SENTRY-6G | Safe recovery when agent execution and network acknowledgment diverge

Shriyam Adya Sharma • Shreyas G